

SIMATS ENGINEERING

ASSESSMENT TOOL – I : SIMULATION TASK

Course Code	ECA1422
Course Name	Embedded Systems
Course Outcome	CO2
Assessment Tool	Simulation Task
Question No.	9
Simulation Task	Construct a simulated Embedded C program for a wearable healthcare device with serial communication and low power operation'
Maximum Marks	30
Bloom's Taxonomy Level	BL3

AIM: To construct a simulated Embedded C program for a wearable healthcare device that integrates serial communication for data transmission and utilizes low-power operation to maximize battery efficiency.

PROBLEM DESCRIPTION:

Wearable healthcare devices require continuous operation while relying on small battery sources. The challenge is to design a firmware architecture that accurately reads physiological data (such as heart rate) and transmits it to an external monitor via serial communication, while spending the majority of its time in a low-power sleep state to conserve energy.

SYSTEM REQUIREMENTS:

- Hardware (Simulated): ATmega328P Microcontroller (Arduino Uno), Potentiometer (for simulating analog heart rate sensor).
- Software: Wokwi Online Simulator, AVR-GCC compiler, Built-in Serial Monitor.

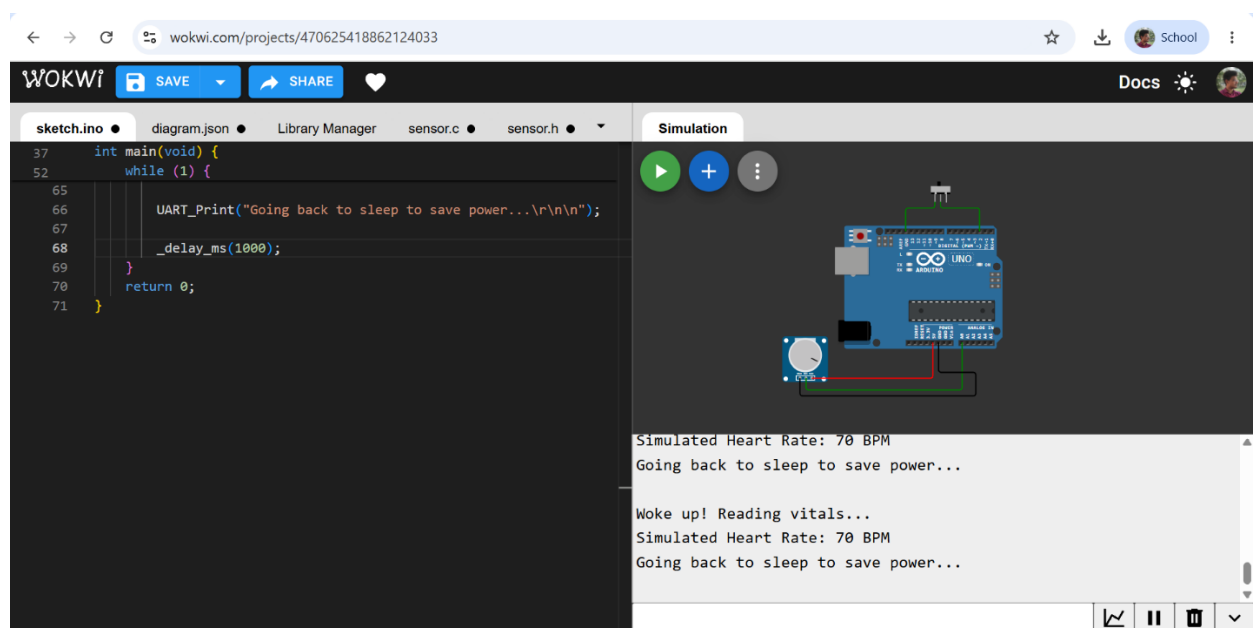
- Protocols: UART for serial communication.

SYSTEM DESIGN:

The system architecture centers around an ATmega328P microcontroller. An analog potentiometer is connected to the ADC pin (A0) to simulate heart rate fluctuations. The microcontroller utilizes an internal Watchdog Timer to trigger wake-up interrupts at regular intervals. Once awake, the system reads the ADC value, scales it to represent Beats Per Minute (BPM), transmits the formatted data string via UART to a serial monitor, and immediately re-enters the `SLEEP_MODE_PWR_DOWN` state to minimize current draw.

ALGORITHM:

1. Initialize UART communication at a 9600 baud rate.
2. Initialize the Analog-to-Digital Converter (ADC).
3. Configure the Watchdog Timer to generate an interrupt every 2 seconds.
4. Enter `SLEEP_MODE_PWR_DOWN`.
5. On Watchdog Interrupt: Wake up the microcontroller.
6. Trigger an ADC conversion on Pin A0 and read the raw value.
7. Map the raw ADC value (0-1023) to a realistic heart rate range (60-140 BPM).
8. Format the BPM data into a string and transmit it via UART.



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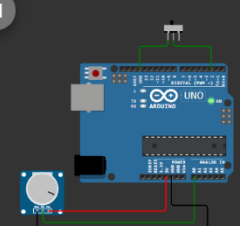
Docs

sketch.ino diagram.json Library Manager sensor.c sensor.h

```
37 int main(void) {
52   while (1) {
65       UART_Print("Going back to sleep to save power...\r\n\r\n");
67       _delay_ms(1000);
68   }
69   return 0;
70 }
71 }
```

Simulation

00:04.451 100%



System Initialized. Entering Low Power Mode...

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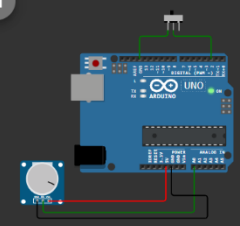
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Simulation

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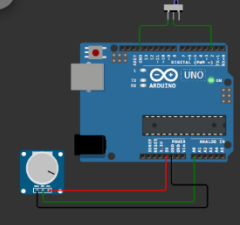
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65       UART_Print("Going back to sleep to save power...\r\n\r\n");
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68   }
69   return 0;
70 }
71 }
```

Simulation

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System Initialized. Entering Low Power Mode...
Woke up! Reading vitals...
Simulated Heart Rate: 70 BPM
Going back to sleep to save power...

Evaluation Rubric (30 Marks)

Criteria	Weightage	Level 4	Level 3	Level 2	Level 1	Marks Awarded
Requirements Analysis & Conceptual Understanding	6	☐ 6	☐ 5	☐ 4	☐ 2–3	
System Design & Simulation Model Development	8	☐ 8	☐ 6	☐ 5	☐ 3–4	
Implementation & Tool Usage	6	☐ 6	☐ 5	☐ 4	☐ 2–3	
Testing, Validation & Result Analysis	6	☐ 6	☐ 5	☐ 4	☐ 2–3	
Documentation	4	☐ 4	☐ 3	☐ 2	☐ 1	
Total	30					_____/30

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: